

PP3S Diary 4

Group E

May 2026

This week, me and Doruk extended the backend stock module by implementing database-based stock trading logic. We added functionality for fetching the latest stock price from the database, as well as basic buy and sell service logic with validation checks for portfolio balance and stock holdings. In addition, trade creation for buy and sell operations was implemented and integrated into the existing backend module structure. We also worked with the team's Git workflow by updating the local dev branch, creating a feature branch, and opening a Pull Request for review. (Mehmet and Doruk)

This week, due to overlapping deadlines with another course assignment, I was not able to make significant implementation progress. However, I continued following the progress of other team members and stayed involved in the ongoing development process. Since the foundational database structure has already been completed, I hope to discuss the next stage of tasks during this week's meeting.

I also spent more time learning about the specific requirements and business logic behind stock-related software. Since I previously had very limited exposure to stock systems, this has been a particularly new and interesting learning experience for me. Moving forward, I plan to build upon the database structure I have already developed and further improve the trading-related logic of the system.

With the midterm presentation approaching, I will also look into ways I can contribute to the preparation work next week. (Yifan)

This week, we discussed in the group meeting the next tasks. I showcased our new frontend and did a progress review on the backend tasks, that my other group members did. David and I then distributed more issues on the Linear Kanban board. I reviewed the PRs, rebased them and fixed the linting errors. The days after the meeting, I finished the UI and deleted all the mocks from the frontend, to begin integrating the real finished modules. (David)

This week, we mainly discussed the general backend structure of our stock exchange project and divided the tasks within the team. Together with Ella, I

will work on the user and portfolio/transaction modules.

We briefly talked about how user accounts, balances, and stock transactions could work and collected initial ideas for connecting users with their portfolios.

Overall, this week was mostly about getting an overview of the project structure and preparing for the implementation phase. (Tekle, Ella)

This week I spent my time implementing the sync service, which periodically pulls in stock data from the external API. This required some restructuring in the general application structure due to the introduction of inter-service dependencies. Due to the Alpha Vantage APIs aggressive rate limits, which were entirely unsuitable for testing, I also implemented a client for the TU stocks API. The most interesting technical part of this work was probably a database row-as-lock mechanism to ensure that only one app container attempts the sync at a time - we don't currently plan on deploying multiple app instances, but it's nice to keep it horizontally scalable, at least in theory. (Moritz)

I spend time reworking the frontend. David raised some interesting thoughts concerning the readability last meeting. That is why I spend this week making mockups which I present to the group next meeting. (Oliver)

This week David and I again reviewed PRs and managed issues through Linear. In addition, I spend a lot of time getting the big picture of the codebase. David and I also created the first mermaid diagram of the architecture. Before we only had a sketch in TLDraw. Then we discussed where we need to consolidate or restructure. (David A)